



Satisfeed Food Delivery Assistant

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INTRODUCTION

This project was developed under the guidance of Dr. Anca and in partnership with client Tim Turner of Satisfeed, a non-profit organization dedicated to fighting food insecurity.

PROBLEM & OBJECTIVE

We analyzed socioeconomic data to find the true drivers of food insecurity, ensuring Satisfeed delivers food to the communities that need it most.

METHODS & TOOLS

1. Pandas
2. Matplotlib
3. Principal Component Analysis (PCA)
4. Standard Scaler
5. K-means Clustering
6. Correlation Analysis

1. Google Colab
2. Jira
3. Github

CONCLUSION

Our analysis notebook transforms raw datasets into built knowledge, equipping the Satisfeed dashboard with the insights needed to automate and optimize high-need food distribution.

DATASETS

Feeding America:

<https://www.feedingamerica.org/research/map-the-meal-gap/by-county>

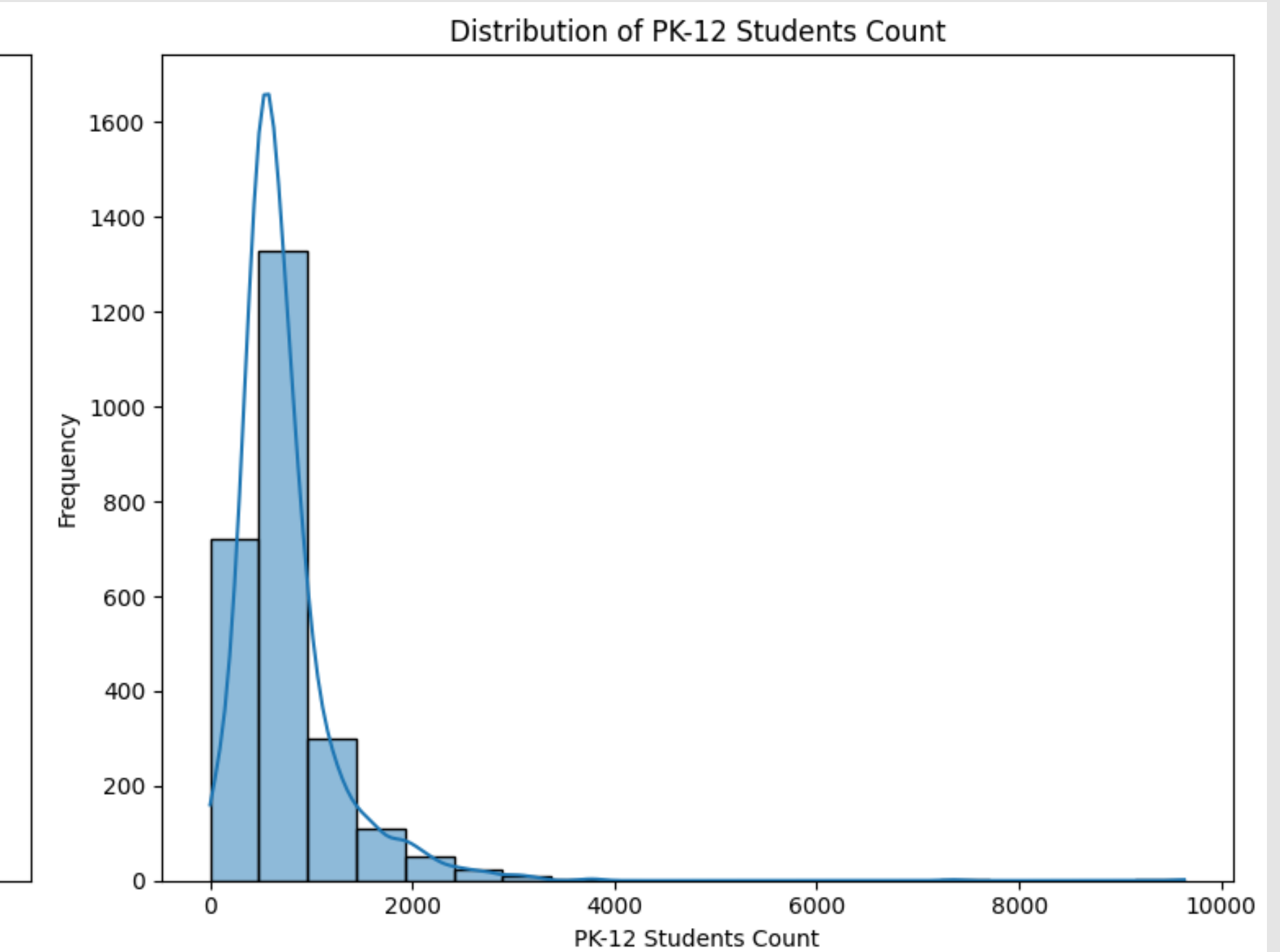
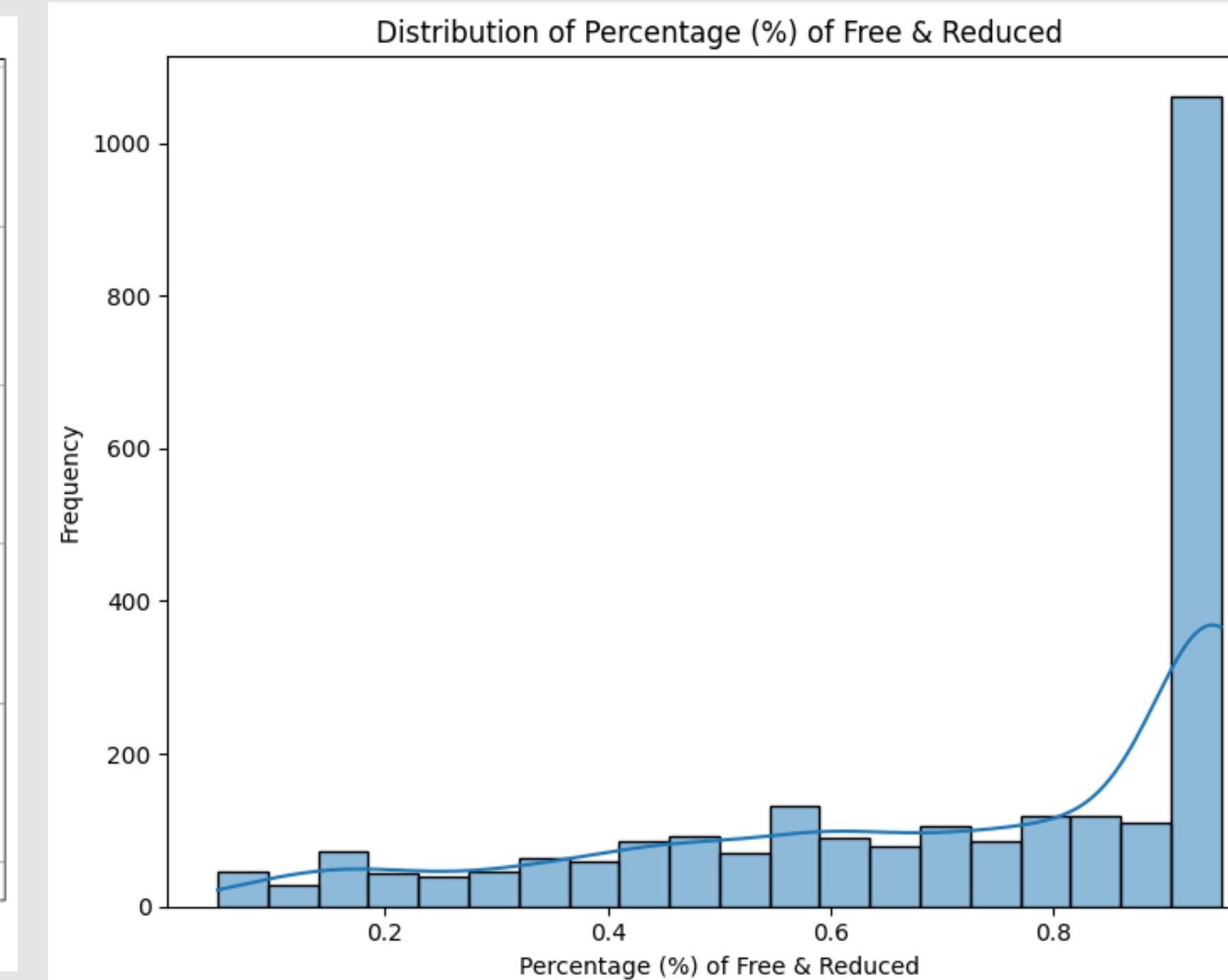
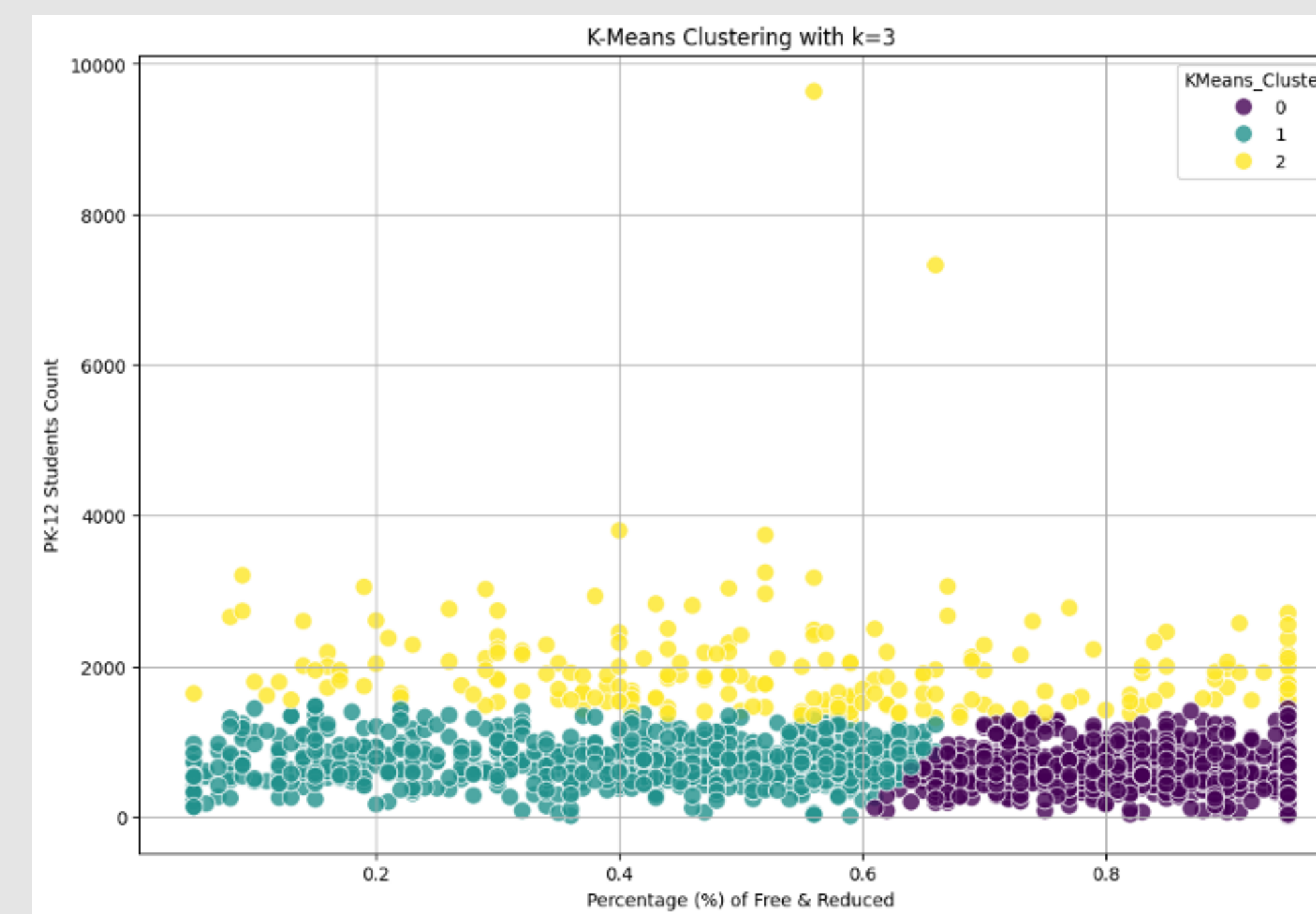
Georgia Dept of Early Care and Learning

<https://www.decal.ga.gov/SFSP/Eligibility.aspx?utm>



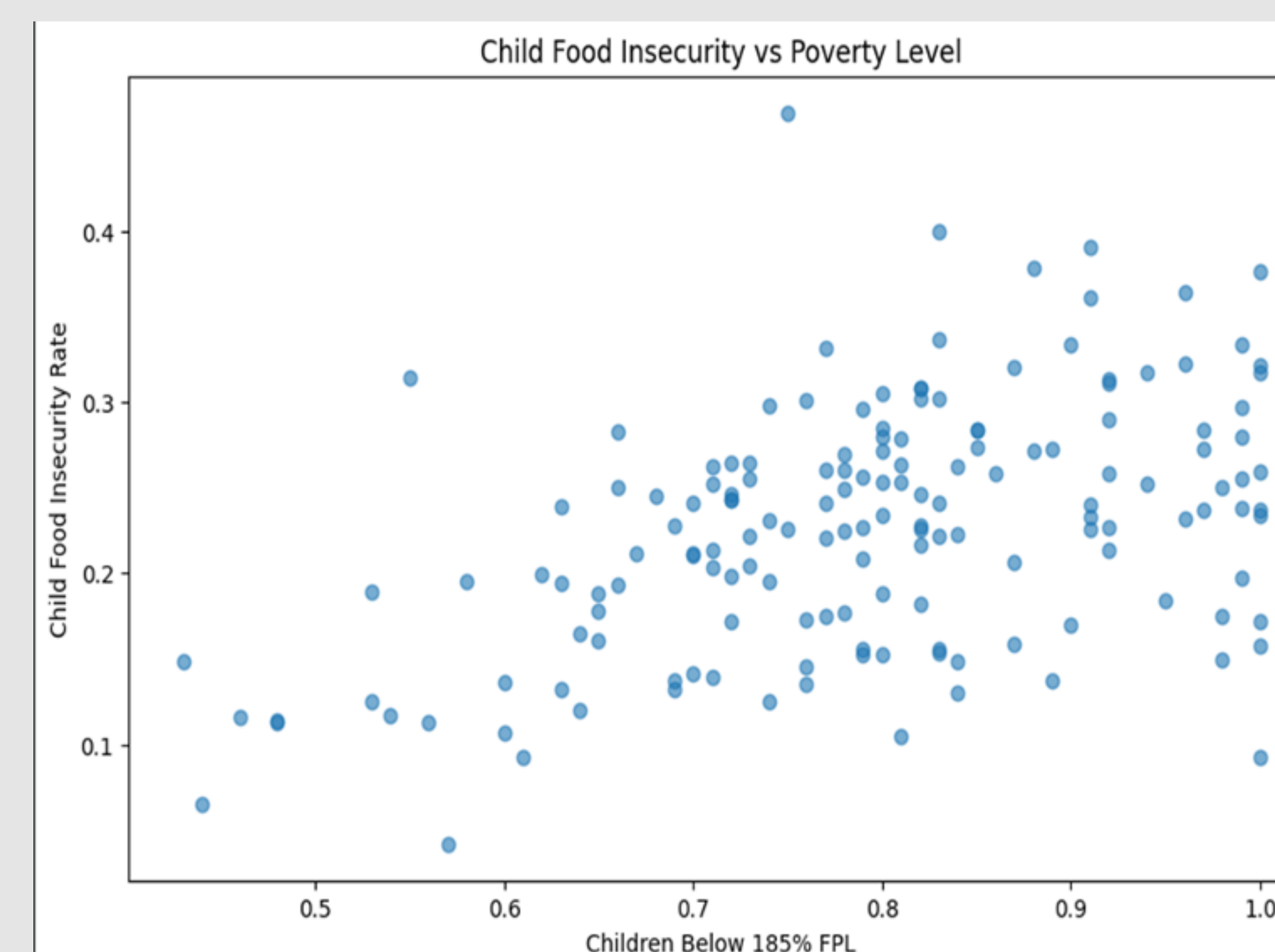
Bigger schools don't mean larger percentage

KMeans_Cluster
0 1593
1 731
2 218

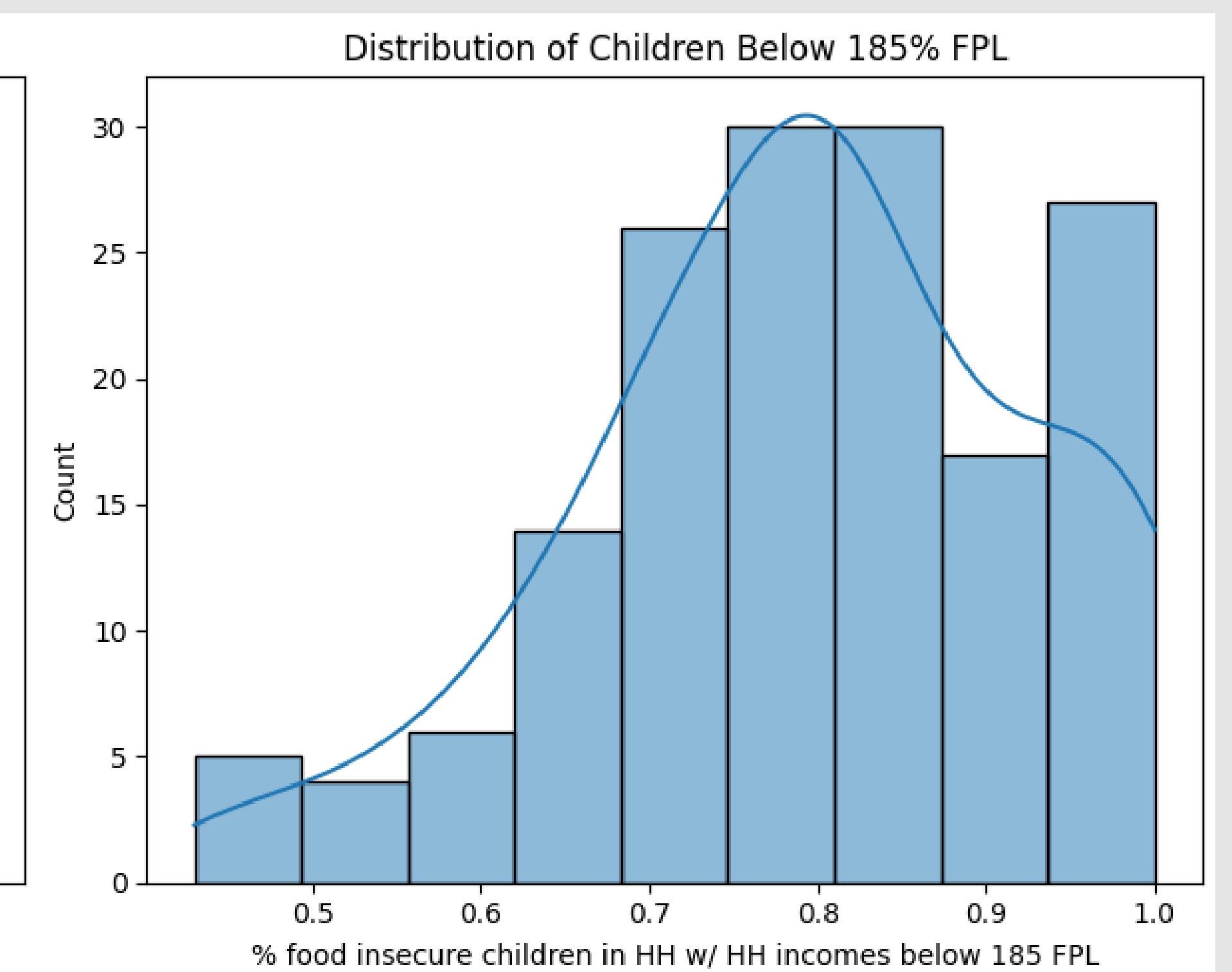
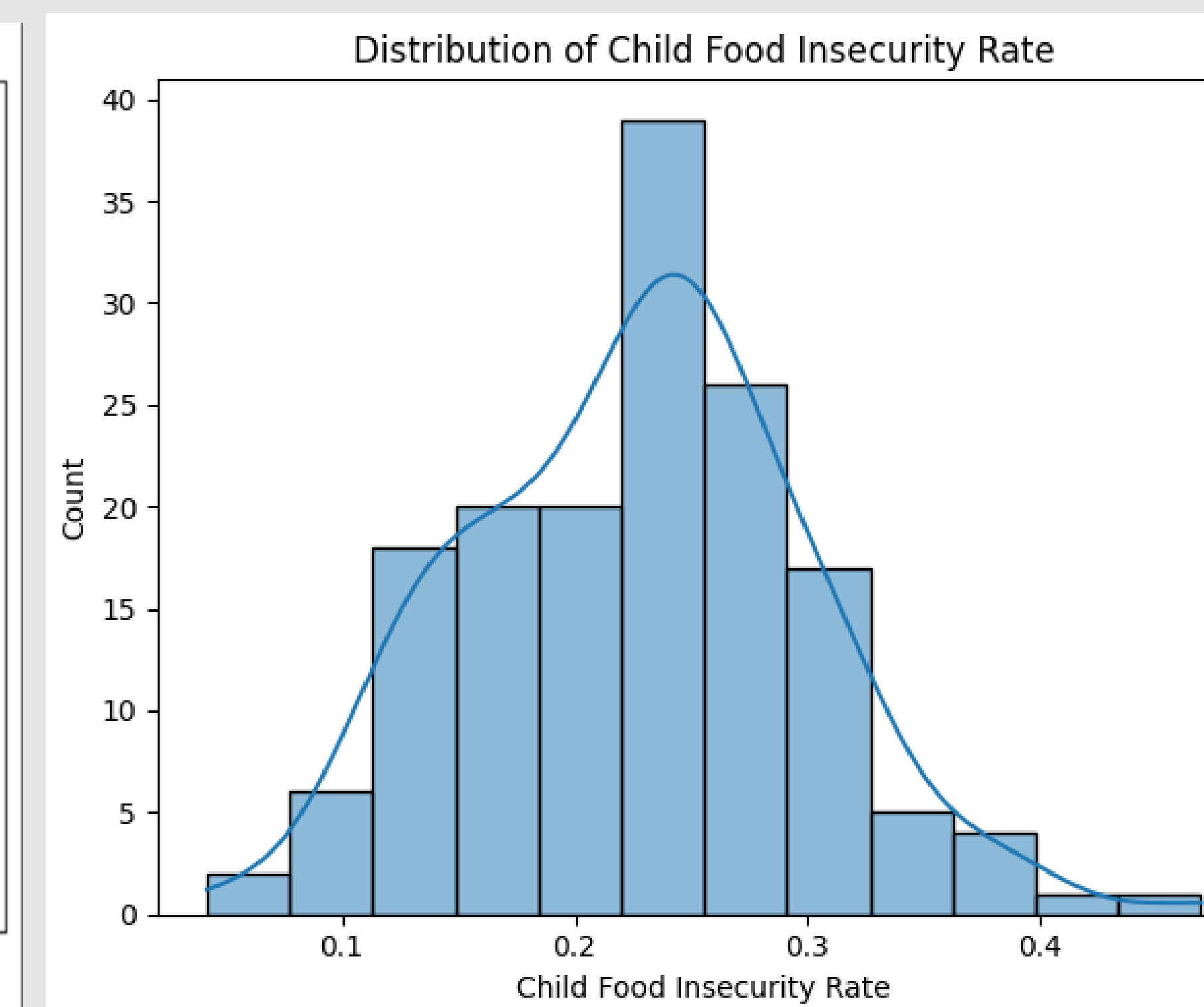


K-means clustering reveals no link between school size and Free & Reduced Lunch rates

Poverty is a key driver of child food insecurity



Correlation Analysis
Correlation factor: 0.45877511392552756



Moderate positive relationship ($r \approx 0.46$): higher poverty leads to higher child food insecurity.

School Population doesn't contribute to high rates of free and reduced

PCA analysis exposes the enrollment fallacy: total school population is statistically disconnected from a campus's actual level of socioeconomic need.

